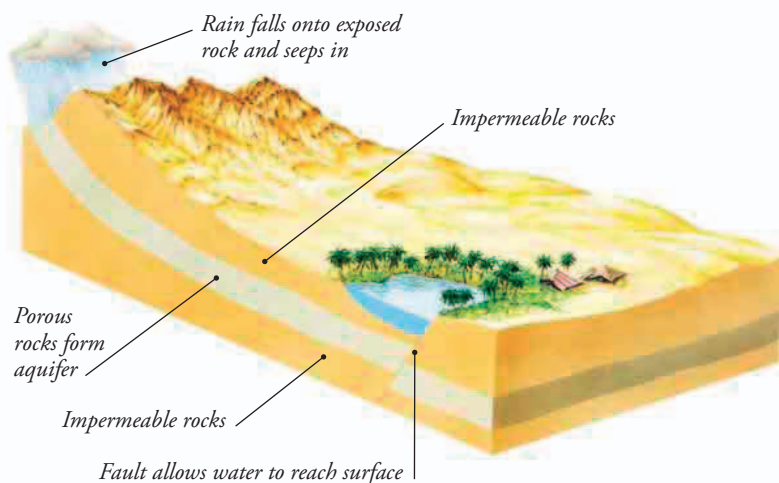


AQUIFERS

Porous rocks that contain water are called aquifers. The water usually flows very slowly downhill over impermeable rock below. If an aquifer is capped with more impermeable rock, it can completely fill with water. However, if there is a fault (a gap or break) in the rock, water can seep to the surface.



SPRINGS

Impermeable rock strata often outcrop (come to the surface) on hillsides, so the water stored above them flows out at this level as a spring. Usually, a whole layer of rock outcrops in this way, and a row of springs marks its position. Springwater is normally very pure because the porous rock it passes through acts as a filter.



ANCIENT WATER

Some deserts conceal vast underground water reservoirs formed thousands of years ago. One beneath the eastern Sahara contains about 36,000 cubic miles (150,000 cubic km) of water. These ancient water reserves are capped by layers of waterproof rock, but wind erosion can strip away the rock to expose the water and create an oasis (below).



PERMAFROST

On Arctic tundra, water seeping into the soil freezes solid. It thaws near the surface in summer, but stays frozen below ground. This permafrost layer stops the summer meltwater from draining away, so it forms vast areas of swamp. Repeated freezing and thawing opens cracks that fill with stones, creating a strange effect called patterned ground.

